$$\left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - 3v_{th}^2 \nabla^2\right) n_e' = \frac{n_0 e^2}{m_e^2 c^2} \nabla^2 (\vec{A}_L \cdot \vec{A}')$$

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \nabla^2 + \omega_{pe}^2\right) \vec{A}' = -\frac{4\pi e^2}{m_e} n_e' \vec{A}_L$$

$$\left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - 3v_{eth}^2 \frac{\partial^2}{\partial x^2}\right) n_e' = \frac{n_0 e^2}{m_e^2 c^2} \frac{\partial^2}{\partial x^2} (\vec{A}_L \cdot \vec{A}')$$

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} + \omega_{pe}^2\right) \vec{A}' = -\frac{4\pi e^2}{m_e} n_e' \vec{A}_L$$

$$\left(-\omega^2 + \omega_{pe}^2 + 3v_{eth}^2 k^2\right) n_e' = \frac{n_0 e^2}{m_e^2 c^2} k^2 (\vec{A}_L \cdot \vec{A}')$$

$$\left(-\omega^2 + c^2 k^2 + \omega_{pe}^2\right) \vec{A}' = -\frac{4\pi e^2}{m_e} n_e' \vec{A}_L$$

$$\vec{A}' = -\left(-\omega^2 + c^2 k^2 + \omega_{pe}^2\right)^{-1} \frac{4\pi e^2}{m_e} n_e' \vec{A}_L$$

$$\left(-\omega^2 + \omega_{pe}^2 + 3v_{eth}^2 k^2\right) \underset{\sim}{n_e'} = \frac{n_0 e^2}{m_e^2 c^2} k^2 \left(-\omega^2 + c^2 k^2 + \omega_{pe}^2\right)^{-1} \frac{4\pi e^2}{m_e} \underset{\sim}{n_e'} A_L^2$$